Exploring Positional Distributions of Labeled Superpixels Within Graph Convolutional Networks for Hyperspectral Image

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***Abstract*— Researchers have been paying more attention to hyperspectral image (HSI) classification based on semi-supervised superpixel-level graph convolutional networks (SGCNs) due to their aggregation ability of rich contextual information. Although these SGCNs achieve good classification performance, the influence of the positional distributions among labeled superpixels has been overlooked. The locations of labeled superpixels, such as located at class boundaries or centers, exert a substantial influence on the final performance. To address this issue, this article proposed a novel graph neural network (GCN) method with the guidance of positional distributions of labeled superpixels, abbreviated as LPDGCN. Specifically, we first propose to utilize the sparse, low-rank as well as feature smoothness restrictions to optimize the initial superpixel graph structure because the connectivity relationships of labeled superpixels located at class boundaries or centers are easily influenced by spectral variation. Second, in order to effectively determine the positional distributions of labeled superpixels and make full use of the position relationships, we propose to utilize the information conflict from the above topology connectivity to determine the positional distributions of labeled superpixels and develop the reweighted strategy to weaken the influence of labeled superpixels located at class boundaries and strengthen the influence of that located at class centers. Finally, we evaluate the LPDGCN method on four public HSI datasets, demonstrating its superiority over other advanced classification methods in terms of three metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC).**

***Index Terms*— Graph convolutional network, hyperspectral image (HSI) classification, positional distribution.**

I. Introduction

HYPERSPECTRAL techniques are increasingly employed in various Earth observation applications owing to advancements in optical technology. Compared with panchromatic and multispectral imagery, hyperspectral images (HSIs) possess high spectral resolution at the nanometer level [1], [2], [3], [4], which can be manifested as subtle differences in the spectral dimension among various objects on the ground, enabling precise characterization of their reflective spectra. As a result, it significantly enhances the capability for object classification [5], [6]. HSI classification has gained significant attention due to its crucial role in remote sensing, environmental monitoring, exploration, agriculture, and many other fields [7], [8], [9], [10]. In general, existing methods can be broadly divided into two categories, including hand-crafted and deep learning-based techniques.

The purpose of HSI classification is to input the spectral features of each training pixel into a trained classification model to determine the corresponding category of testing pixels. In the early stages, hand-crafted feature extraction techniques garnered more attention. For instance, the principal component analysis (PCA) [11], [12] and independent component analysis (ICA) [13] were proposed to extract effective feature information, aiming to extract the primary features from the data, reduce redundant interclass information or reduce the dimensionality of the data. However, due to the substantial variability in the spectral feature space caused by various factors, classification results based solely on spectral information are insufficient to support accurate object classification. Consequently, researchers have proposed a series of methods that combine spectral and spatial information to achieve effective classification, such as the spatial–spectral Markov random field (MRF) [14] method. Although the aforementioned hand-crafted feature extraction methods achieve good classification results, they still have limitations, particularly in performing effective end-to-end training within a unified framework while considering downstream tasks.

In recent years, there has been rapid development in deep learning techniques [15], [16], with convolutional neural networks (CNNs) [17] being notable representatives that have demonstrated remarkable progress in HSI classification. As powerful image processing tools, CNNs are widely applied to HSI classification, facilitating automatic feature learning through stacked layers of convolutional, pooling, and fully connected layers. This enables CNNs to extract rich spectral and spatial features from hyperspectral data, thereby improving classification results. Generally, CNN-based methods are primarily divided into spectral CNNs, spatial CNNs, and spatial–spectral CNNs. Spectral CNN

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mainly refers to 1-D CNN, which performs 1-D feature extraction of spectral bands. Specifically, the spectrum of each pixel is treated as a sequence, and 1-D CNN slides 1-D convolution kernels along the spectral dimension to extract spectral features [18]. However, 1-D CNN has limitations in capturing spatial relationships between pixels. To address this problem, He et al. [19] proposed a 2-D CNN, also known as spatial CNN. The core principle of 2-D CNNs in HSI classification lies in their ability to perform 2-D feature extraction in the spatial dimension, capturing spatial relationships around pixels. Jia et al. [20] proposed an advanced 2-D CNN, which can utilize the inherent spatial geometric information more effectively. Furthermore, in order to better utilize both the spatial and spectral information of HSIs, Chen et al. [21] proposed 3-D CNNs, which conduct comprehensive 3-D spatial–spectral feature extraction. However, 3-D CNNs suffer from high computational complexity due to large parameter sizes. In response, Zhang et al. [22] proposed a dual-branch network that splits the 3-D CNN into two branches: one for spectral information extraction using 1-D CNNs, and the other for managing spatial information using 2-D CNNs, thereby mitigating computational complexity to some extent. In addition, there are some CNN models have been refined from alternative perspectives. For example, deep pyramidal residual networks (DPRNs) [23] proposed a residual-based approach, which can gradually increase the feature map dimension at all convolutional layers. Furthermore, the dual-view spectral and global spatial feature fusion network (DSGSF) [24] proposed a dual-view spectral feature aggregation model, extracting spatial–spectral features with stronger discriminative power.

However, due to the complex spatial structure and irregular distribution in HSI data, CNNs might exhibit suboptimal performance when handling irregular land cover distribution. Besides, the limited availability of training samples makes CNNs susceptible to feature overfitting in scenarios with a scarcity of samples. In such cases, graph neural networks (GCNs) often excel in handling irregular data distributions through flexible graph convolutions using a message-passing paradigm with semi-supervised learning way [25], meeting the basic requirement of HSI land cover classification and achieving good classification results. Due to the original GCN only utilizing spectral feature information, some researchers have proposed GCN-based methods by combining spatial and spectral information. These existing methods are generally divided into local spatial–spectral GCNs and global spatial–spectral GCNs. Representatives of local spatial–spectral GCNs are S^2GCN [26], miniGCN [27], and Funet-C [27] methods. The S^2GCN method combined inherent spectral information with local spatial window information for HSI classification. However, it struggles with larger input graph structures in HSI, prompting the development of miniGCN and Funet-C, which reduces complexity by partitioning large graph structures into smaller graphs. In addition, a fast dynamic graph convolutional network and CNN parallel network (FDGC) was proposed for HSI classification [28], which enhanced the classification performance and improved computational efficiency through flattening and pooling operations. While these methods achieve local feature smoothing through graph convolution, they may be limited in their ability to comprehensively capture contextual information. Consequently, researchers have proposed a global contextual property of GCN-based methods, which aims at pursuing the global consistency of intraclass spectral information [29], [30]. For example, some researchers have paid more attention to the pixel-level global graph structure [31]. However, these approaches may struggle with large-scale HSI due to pixel-level exponential neighborhood expansion. To address this challenge, researchers have proposed superpixel-level global GCN-based methods [32], such as the multiscale dynamic graph convolutional network (MDGCN) [33], CNN-enhanced graph convolutional network with pixel and superpixel-level feature fusion (CEGCN) [34], and attention multihop graph and multiscale convolutional fusion network (AMGCFN) [35] methods. These methods utilize superpixel-level graph structures for neighbor information aggregation by a graph convolution operation.

In fact, some methods based on local spatial–spectral GCNs and global spatial–spectral GCNs achieve good classification results from the perspective of pixel-level GCNs, but superpixel-level GCNs have lower computational complexity and capture contextual semantic information, alleviating spectral variation in HSI. Meanwhile, the aforementioned superpixel-level GCN-based methods ignore the influence of position distributions of labeled superpixels in superpixel-level GCN-based methods. For these labeled superpixels located at class boundaries, their connectivity relationships with other superpixels are more prone to being influenced by other classes due to the similarity of interclass spectral features, leading to the deviation of the classifier's decision. Conversely, for these labeled superpixels located at class centers, they are more likely to establish relationships with a greater number of homogenous superpixels within the same class, potentially improving the discriminant ability of the classifier decision. However, due to the spectral variation of intraclass information, the positive effects for samples located at class centers may be somewhat constrained, while the negative effects for samples located at the class boundaries may be exacerbated. To this end, we conducted some experiments on the Indian Pines dataset to explore the impact of the positional distribution of labeled superpixels based on GCNs in Fig. 1. The reasons for selecting the Indian Pines dataset are from its typical spectral variability and position-imbalanced characteristics, leading to severe confusion in the structural relationships between labeled and unlabeled superpixels, and exacerbating the influence of the position distribution of labeled superpixels. As shown in Fig. 1(a), when the positions of the labeled superpixels are more evenly distributed, the results are significantly better compared to situations where most labeled superpixels are concentrated at class centers or class boundaries. Then, we can draw a conclusion that different positional distributions of labeled superpixels have a significant impact on the final classification performance. Next, another issue emerges: in superpixel-based GCN methods, aside from the position relationships of superpixels affecting classification performance, does the adjacency connectivity

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of superpixels at different locations also have a significant impact on the final classification results? Therefore, the quality of the superpixel-level graph structure is also explored via related experiments in order to seek more solutions for the influence of positional distributions. Specific experiments can be observed in Fig. 1(b), which demonstrates that different graph structures of the same uniform distribution with superpixels correspond to different classification performances because graph connectivity relationships of superpixels located at class boundaries or class centers are easily influenced by the complex characteristics of HSIs. Thus, the quality of superpixel-level graph structures with different position distributions requires additional attention.

In order to address these issues, this article proposed the LPDGCN method for HSI. For the first issue, we develop a position-reweighted training strategy to alleviate the influence of labeled superpixels in different positional distributions because the labeled superpixels are randomly distributed at class boundaries or centers in practical scenarios. Next, for the second issue, we develop an optimized strategy with graph structure to reconstruct the new graph structure to alleviate the influence of spectral variation. The ultimate goal is to address the influence of the positional distributions of labeled superpixels in GCN-based classification performance. The optimization of the graph structure aims at accurately determining the positional relationships of labeled superpixels. Consequently, the order of the following study cannot be reversed. The graph structure can be first optimized, followed by the determination of positional distributions of labeled superpixels and their consequent effects in the GCN training strategy. In summary, the contributions of this article are concluded as follows.

1) This article proposed the sparse and low-rank strategies to remove the interclass connections of graph structure, as well as the feature smoothness strategy to improve the tightness of the graph structure due to the intraclass spectral variation. The above strategies optimized the quality of graph structure between labeled and unlabeled superpixels.
2) This article proposed to utilize the information conflict from the topology graph connectivity to determine the position distributions of labeled superpixels and develop a position-reweighted strategy to weaken the influence of labeled superpixels at class boundaries and strengthen the effect of labeled superpixels at class centers.
3) This article proposed the LPDGCN method to enhance the model's generalization ability through the reconstruction of the graph structure and the position-reweighted strategy to labeled superpixels. Comprehensive experiments demonstrated that the LPDGCN method achieved better classification performance compared to some popular methods.

The remainder of this article is organized as follows. Section II introduces the details of the LPDGCN method. Section III reports the extensive experimental results and analyses. Finally, conclusions are drawn in Section IV.

II. Proposed Method

A. Framework of the Proposed Method

The overall framework of the proposed LPDGCN is illustrated in Fig. 2. First, the SLIC [36] method was applied to initial HSIs to obtain the results of superpixel segmentations for reducing computational complexity as well as capturing multiscale information. Then, we propose to utilize the sparse, low-rank as well as feature smoothness restrictions to alleviate the influence of spectral feature similarity of interclass samples and spectral feature variation of intraclass samples, obtaining a qualified graph structure. Next, we propose to utilize the information conflict from the above topology graph structure to determine the position distributions of labeled superpixels and utilize a reweighted strategy to weaken the influence of labeled superpixels at class boundaries and strengthen the influence of labeled superpixels at class centers. Finally, the LPDGCN is able to address the above questions brought by position distributions of labeled superpixels, achieving accurate model decisions.

B. Optimization of Graph Structure

In order to reduce the computational burden of GCNs, we use the superpixels as graph nodes to construct the graph structure. In fact, the connectivity relationships of superpixels located at class boundary positions and class centers are easily influenced by interclass similarity and intraclass variation of spectral information, resulting in redundant and inaccurate adjacency relations. Therefore, we proposed to utilize the sparse, low-rank as well as feature smoothness restrictions to remove the redundant connections and improve the tightness of the graph structure. Specific details are described as follows. Considering that labeled superpixels located at class boundaries are more likely to be affected by other superpixels from other classes, the sparse constraint is first applied to the given graph structure for removing interclass connectivities. The mathematical expression can be written as

$$L_1 = \|\boldsymbol{P} - \boldsymbol{A}\|_F^2 + \lambda_1 \|\boldsymbol{P}\|_1 \tag{1}$$

where $\boldsymbol{A}$ denotes the initially given graph structure and $\boldsymbol{P}$ denotes the reconstructed graph structure. The first term $\|\boldsymbol{P} - \boldsymbol{A}\|_F^2$ is designed to constrain the reconstructed graph structure $\boldsymbol{P}$ so that it cannot completely escape the building of initial graph structure $\boldsymbol{A}$ because the expert knowledge is also reasonable to a certain degree. The second term $\|\boldsymbol{P}\|_1$ denotes the restriction of the l_1-norm for the reconstructed matrix $\boldsymbol{P}$, where this restriction aims at removing the interclass adjacency relationships of labeled superpixels located at class boundaries. Furthermore, considering that a few interclass adjacency relationships are hard to remove, leading to high-rank characteristics of the adjacency matrix due to its severe spectral similarity of interclass information, the low-rank restriction is further applied to the given graph structure to preserve the subspace property of HSI. A specific mathematical formula can be written as

$$L_1 = \|\boldsymbol{P} - \boldsymbol{A}\|_F^2 + \lambda_1 \|\boldsymbol{P}\|_1 + \lambda_2 \|\boldsymbol{P}\|_* \tag{2}$$

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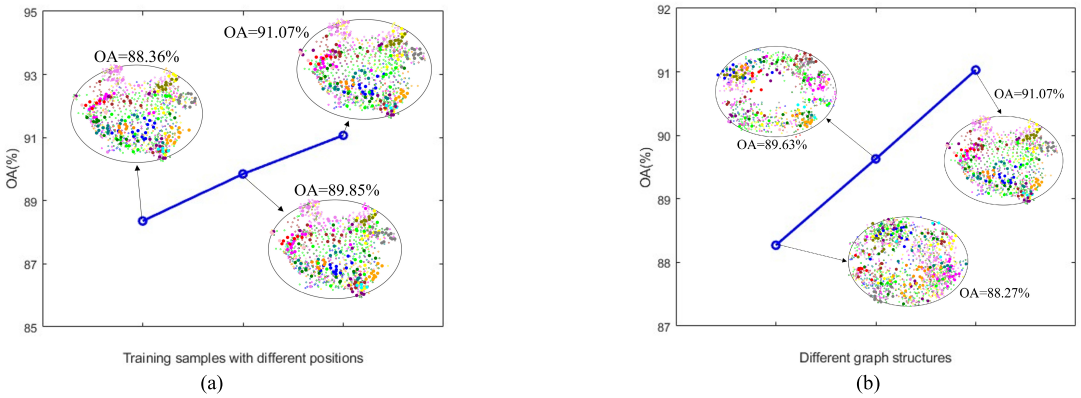

Fig. 1. (a) Classification results under different positions of labeled superpixels. (b) Classification results with different graph structures in same positions of labeled superpixels. The bigger solid nodes represent the labeled samples. Experimental results demonstrate that identifying positional distributions and improving graph structure quality of labeled superpixels corresponding to unlabeled superpixels are important to enhance the final performance.

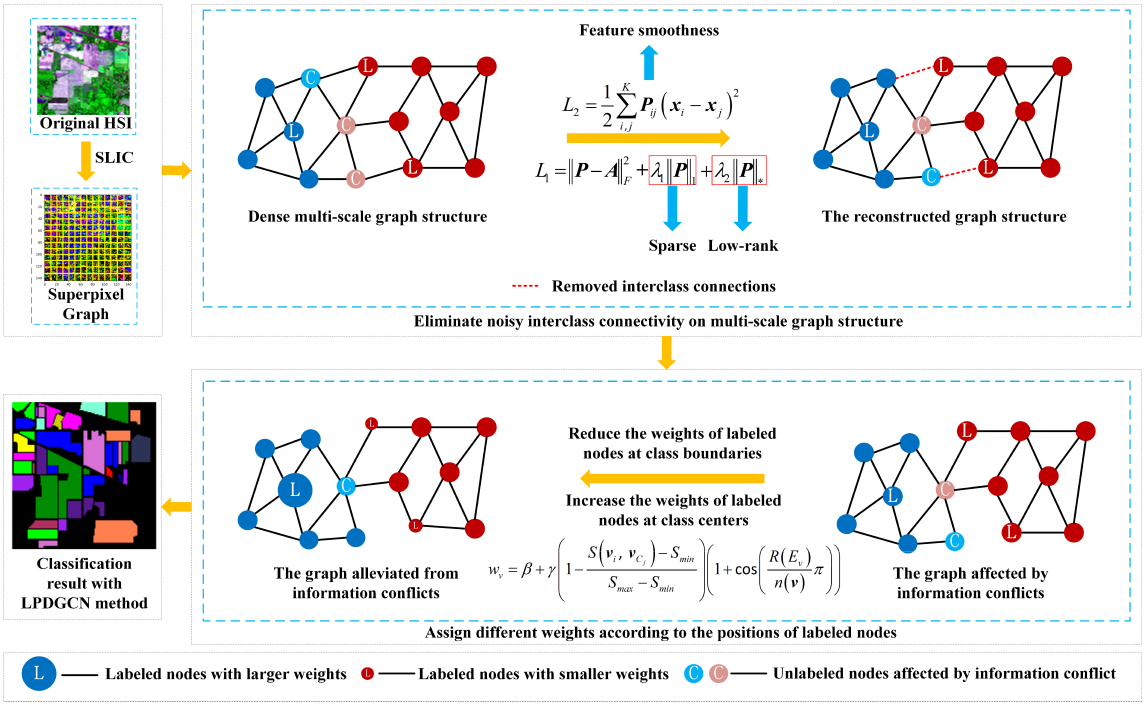

Fig. 2. Pipeline of the proposed LPDGCN method. The entire process begins with superpixel segmentation, followed by the construction of a multiscale graph. Next, multiscale graphs are reconstructed, with red dashed edges indicating interclass connections eliminated during the reconstruction process. Finally, different weights are assigned to labeled nodes according to different topological positions. The colors of different circles indicate different categories.

where the third term $\|\boldsymbol{P}\|_*$ denotes the restriction of the nuclear norm of the matrix $\boldsymbol{P}$ replacing the low-rank constraint due to its NP-hard problem [37]. This restriction aims at removing adjacency relationships of interclass information resulting from severe spectral confusion. The parameters λ_1 and λ_2 are hyperparameters controlling the weights of the sparse norm and nuclear norm, respectively. After obtaining the optimized graph structure via sparse and low-rank restrictions, we hope the optimized graph feedback to the features, making the spectral feature of intraclass superpixels smoothing and improving the tightness of the graph structure. Therefore, the newly obtained adjacency matrix with sparse and low-rank restrictions is feedback to improve the feature smoothness and structure tightness of superpixels located at class centers. The feature smoothness can be written as

$$L_2 = \frac{1}{2}\sum_{i,j} \boldsymbol{P}_{ij}\left(\boldsymbol{x}_i - \boldsymbol{x}_j\right)^2 \tag{3}$$

where $\boldsymbol{P}_{ij}$ represents the adjacency relationships between nodes i and j, and $(\boldsymbol{x}_i - \boldsymbol{x}_j)^2$ represents the feature smoothness degree between any two superpixels. Inspired by Laplacian smoothing and using symmetric normalization techniques [38], we can express L_2 in the following form:

$$L_2 = \mathrm{tr}\left(\boldsymbol{X}^{\mathrm{T}}\hat{\boldsymbol{L}}\boldsymbol{X}\right) = \frac{1}{2}\sum_{i,j}^{K} \boldsymbol{P}_{ij}\left(\frac{\boldsymbol{x}_i}{\sqrt{d_i}} - \frac{\boldsymbol{x}_j}{\sqrt{d_j}}\right)^2 \tag{4}$$

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where $\hat{\boldsymbol{L}} = \boldsymbol{D}^{-1/2}\boldsymbol{L}\boldsymbol{D}^{-1/2}$ and d_i and d_j represent the degree of nodes. In order to construct the graph structure equipped with sparse and low-rank properties and feature smoothness, the loss function can be expressed as

$$L_0 = L_1 + \lambda_3 L_2 = \|\boldsymbol{P} - \boldsymbol{A}\|_F^2 + \lambda_1\|\boldsymbol{P}\|_1 + \lambda_2\|\boldsymbol{P}\|_* + \lambda_3 \mathrm{tr}\left(\boldsymbol{X}^{\mathrm{T}}\hat{\boldsymbol{L}}\boldsymbol{X}\right) \tag{5}$$

where λ_3 is a hyperparameter that controls the importance of feature smoothness. In fact, if the superpixels are connected with large feature differences, the corresponding edges in the new adjacency matrix $\boldsymbol{P}$ will gradually be eliminated in the reconstruction due to the three restrictions. Under the constraints of these conditions, the new adjacency matrix is continuously updated during the computation process.

C. Position-Reweighted Strategy of Labeled Superpixels

After obtaining the optimized graph structure, the next challenge is to determine the positions of labeled superpixels and evaluate the impact of labeled superpixels located at class boundaries and class centers. To tackle this issue, we utilize topology graph connectivity to determine the position distribution of labeled superpixels instead of using spectral feature differences due to common spectral feature variation. In fact, the above sparse and low-rank as well as feature smoothness restrictions have optimized the given graph structure to remove interclass redundant connections and improve the feature smoothness to a certain degree. However, the interclass connections of superpixels located at class boundaries inevitably exist due to the complex spectral characteristics, it is hard to completely remove the interclass connections of superpixels.

Thus, the proposed basic assumption is that the connectivity relationships of labeled superpixels located at class boundaries are likely to be established with superpixels from other classes, and the connectivity relationships of labeled superpixels located at class centers are likely to be established with superpixels from the same class. In other words, there will be likely to encounter information conflict when labeled superpixels are located at class boundaries. The information conflict represents the magnitude of the influence of labeled nodes on other nodes, which can be called as influence distribution of labeled nodes. The greater the overall influence for other nodes, the stronger the information conflict. As a result, we propose to determine the positional distributions of labeled superpixels by calculating their information conflicts from the connectivity of the graph structure. The larger the information conflict, the closer the superpixel is to the class boundaries. In order to compute the information conflicts of labeled superpixels, we can first determine the influence distribution of labeled nodes by using the personalized pagerank (PPR) [39] matrix. The PPR algorithm makes the influence distribution more concrete by computing the probability of labeled superpixels randomly walking to other superpixels. The PPR algorithm is applied to obtain the influence distribution between superpixels. A specific formula can be written as follows:

$$\boldsymbol{P}_{\mathrm{ppr}} = \alpha\left(\boldsymbol{I} - (1-\alpha)\hat{\boldsymbol{P}}\right)^{-1} \tag{6}$$

where $\hat{\boldsymbol{P}} = \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{P}}\tilde{\boldsymbol{D}}^{-1/2}$, $\tilde{\boldsymbol{P}} = \boldsymbol{P} + \boldsymbol{I}$, and the hyperparameter α represents the restart probability. The influence distribution exerted by each node outward can be represented as $\boldsymbol{P}_{\mathrm{ppr}}$ [40]. The greater the influence distribution of a node, the larger the information conflict it can be considered to have, indicating that the node is closer to the class boundary. Then, the approximate positions of labeled superpixels within the class can be determined by the graph connectivity. Therefore, in order to quantify information conflicts, we define the information conflict value of each labeled superpixel by calculating the expectation of the influence distribution

$$E_v = \left(\sum_{j\in[1,k], j\neq g_v} \frac{1}{|C_j|}\sum_{i\in C_j}\boldsymbol{P}_{\boldsymbol{v}_i}\right) \tag{7}$$

where the PPR probability vector of node $\boldsymbol{v_i}$ is represented as $\boldsymbol{P}_{\boldsymbol{v}_i}$, g_v is the ground truth of nodes, and $1/|C_j|$ is the class weight designed to mitigate the impact of class imbalance. The larger the mathematical expectation E_v, the greater the overall information conflict of a node in the graph structure, indicating its proximity to the class boundary. By calculating the expected value E_v of the overall influence distribution, we can transform the probability matrix into a specific numerical value for subsequent weight calculation. Conversely, a smaller expectation indicates a smaller overall information conflict of a node, signifying its proximity to the class center. Equation (7) quantifies the information conflicts and approximates the positions of labeled superpixels.

Next, another question arises how do these superpixels in different locations play different guiding roles in decision boundaries. Intuitively, the influence of labeled superpixels located at class boundaries should be weakened in the process of model training in order to reduce the influence of nodes from other classes in decision boundaries. Also, the influence of labeled superpixels located at class centers should be strengthened in the process of model training in order to improve the influence of labeled superpixels in decision boundaries. As a result, using the information conflicts value, that is, the expectation of each labeled node, we can calculate and assign the weight to each labeled node as follows:

$$w_v = \beta + \gamma\left(1 - \frac{S\left(\boldsymbol{v}_i, \boldsymbol{v}_{C_j}\right) - S_{\min}}{S_{\max} - S_{\min}}\right) \times \left(1 + \cos\left(\frac{R(E_v)}{n(\boldsymbol{v})}\pi\right)\right) \tag{8}$$

where $S(\boldsymbol{v}_i, \boldsymbol{v}_{C_j})$ denotes the spatial distance from labeled node v_i to its class center; $S_{\max}$ and $S_{\min}$ represent the maximum and minimum distance, respectively; $R(E_v)$ denotes the sorting of expected value in ascending order; $n(\boldsymbol{v})$ denotes the number of labeled nodes; and β and γ are hyperparameters, which control the influence of the weight. The specific expression $(1 - (S(\boldsymbol{v}_i, \boldsymbol{v}_{C_j}) - S_{\min})/(S_{\max} - S_{\min}))$ aims at improving the influence of labeled superpixels located at the class center assigning relatively larger weights to nodes closer to the class center, and reducing the influence of labeled superpixels located at the class boundary assigning relatively smaller weights to nodes farther away. $(1 + \cos((R(E_v)/n(\boldsymbol{v}))\pi))$

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denotes the cosine annealing formula results in small weights for labeled nodes with large information conflicts and large weights for labeled nodes with small information conflicts, which means decreasing the weight of the class boundary and increasing the weight of the central node. As a result, we can assign reasonable weights to labeled nodes under the joint control of spatial distance and information conflicts. The position of labeled nodes closer to the class center is assigned higher weights, while labeled nodes closer to the class boundary are assigned lower weights. As a result, the final loss function is as follows:

$$L_w = -\sum_{i \in \boldsymbol{V}_l} \sum_{j=1}^{C} w_v \boldsymbol{Y}_{ij} \ln \boldsymbol{Z}_{ij} \tag{9}$$

where $\boldsymbol{V}_l$ denotes the corresponding indices of labeled samples, $\boldsymbol{Y}$ denotes the corresponding label indicator, C is the number of all classes, which is also the final dimension of the output features; and w_v is the information conflicts' weight of each labeled node. In addition, our overall approach is mainly divided into two parts. It is worth noting that this method can also be modified in the process by integrating the step of reconstructing the graph into the overall loss function. This allows for dynamic updating of the parameters in the GCN model during the feature information aggregation process, and it can be represented as follows:

$$L = \lambda L_w + L_0 \tag{10}$$

where L_w is the loss function of LPDGCN via (9) and L_0 represents the loss of removing redundant connections via (5). By this means, the information in the GCN loss function can also guide the graph reconstruction process, meaning that the reconstructed adjacency matrix $\boldsymbol{P}$ will have an additional constraint. Algorithm 1 describes the entire process of LPDGCN.

Algorithm 1 LPDGCN Model for HSI Classification

Input: Given the HSI feature matrix $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) \in \mathbb{R}^{m \times d}$ and the initial graph $\mathbf{G}_i$ from datasets.

1) Obtain a superpixel graph $\mathbf{G}$ through the SLIC algorithm for superpixel segmentation by $\mathbf{G}_i$
2) Obtain the $\boldsymbol{w}_v$ of all labeled samples via equation (8)
3) **Training procedure (Epochs):**
   For $t = 1$: epochs:
   Rebuild new clean adjacency matrixs at all scales to eliminate redundant connections via equation (5) and update parameters via equation (10).
   End For
4) Predict the labels of testing samples.

Output: Classification results of the testing samples.

D. Algorithm Optimization

The constraint L_w in (10) further complicates the reconstruction of $\boldsymbol{P}$. Therefore, an alternating optimization scheme can be employed to update $\boldsymbol{P}$. First, split the equation into two parts

$$L_0 = L(\boldsymbol{P}, \boldsymbol{A}) + \lambda_1 \|\boldsymbol{P}\|_1 + \lambda_2 \|\boldsymbol{P}\|_* \tag{11}$$

$$L(\boldsymbol{P}, \boldsymbol{A}) = \|\boldsymbol{P} - \boldsymbol{A}\|_F^2 + \lambda L_w + \lambda_3 L_2. \tag{12}$$

For $L(\boldsymbol{P}, \boldsymbol{A})$, we can use a forward–backward splitting method [41] that alternates between gradient descent and proximal steps

$$\boldsymbol{P}^{(k)} = \text{prox}_{\eta R}\left(\boldsymbol{P}^{(k-1)} - \eta \nabla_{\boldsymbol{P}} L(\boldsymbol{P}, \boldsymbol{A})\right) \tag{13}$$

where η represents the learning rate and prox_R is the proximal operator

$$\text{prox}_R(\boldsymbol{Z}) = \text{argmin}_{\boldsymbol{P}} \frac{1}{2}\|\boldsymbol{P} - \boldsymbol{Z}\|_F^2 + R(\boldsymbol{P}). \tag{14}$$

The proximal operator of $\lambda_1 \|\boldsymbol{P}\|_1$ and $\lambda_2 \|\boldsymbol{P}\|_*$ can be represented as [42], [43]

$$\text{prox}_{\lambda_1 \|\cdot\|_1}(\boldsymbol{Z}) = \text{sgn}(\boldsymbol{Z}) \odot \left(|\boldsymbol{Z}| - \lambda_1\right)_+ \tag{15}$$

$$\text{prox}_{\lambda_2 \|\cdot\|_*}(\boldsymbol{Z}) = \boldsymbol{U} \cdot \text{diag}\left(\left(\sigma_i - \lambda_2\right)_+\right)_i \cdot \boldsymbol{V}^{\mathrm{T}} \tag{16}$$

where $\boldsymbol{Z} = \boldsymbol{U} \cdot \text{diag}(\sigma_1, \ldots, \sigma_n) \cdot \boldsymbol{V}^{\mathrm{T}}$ denotes the singular value decomposition of $\boldsymbol{Z}$, which is used to approximate the effect of the nuclear norm. The symbol sgn denotes the sign function, which returns 1 for positive values, -1 for negative values, and 0 for zero in a matrix. The symbol $(|\boldsymbol{Z}| - \lambda_1)_+$ involves scaling the original matrix values using the parameter λ_1 and setting all negative values in the result, corresponding to the original zero values in the matrix. These are then combined using the Hadamard product to approximate the effect of the 1-norm. Richard et al. [44] introduce the incremental proximal descent method based on the introduced proximal operators. With the above equation, we can iterate to update $\boldsymbol{P}$ in the following cyclic manner:

$$\begin{cases} \boldsymbol{P}^{(k)} = \boldsymbol{P}^{(k-1)} - \eta \nabla_{\boldsymbol{P}}(L(\boldsymbol{P}, \boldsymbol{A})) \\ \boldsymbol{P}^{(k)} = \text{prox}_{\eta\lambda_1 \|\cdot\|_1}\left(\boldsymbol{P}^{(k)}\right) \\ \boldsymbol{P}^{(k)} = \text{prox}_{\eta\lambda_2 \|\cdot\|_*}\left(\boldsymbol{P}^{(k)}\right). \end{cases} \tag{17}$$

After the updating process, we can obtain the optimized matrix $\boldsymbol{P}$ according to different constraints to satisfy the conditions.

III. Experimental Results and Analysis

In this section, to demonstrate the superior performance of the LPDGCN model, we conducted a series of experiments on three real HSI datasets. First, we compared it with some state-of-the-art models in the field. Second, we conducted ablation experiments on the two main methods of the LPDGCN model to demonstrate the effectiveness of the methods.

A. Dataset Description

This study employed four typical HSI datasets, namely, Indian Pines, Pavia University (PaviaU), Houston University 2013 (Houston), and WHU-Hi-HanChuan.

1) The Indian Pines dataset is a well-known hyperspectral dataset that represents a rural area in Indiana, USA. It was collected by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and contains spectral information from 145 × 145 pixels, with each pixel having 224 spectral bands. This dataset is widely used in remote sensing research and serves as a benchmark for HSI classification and analysis tasks due to its complexity and real-world relevance. Fig. 3(a) and (b)

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Fig. 3. Indian Pines data: (a) false-color image (bands 20, 69, and 178 are red, green, and blue, respectively) and (b) ground-truth image.

presents the false-color image along with its corresponding ground truth.

2) PaviaU is another widely used hyperspectral dataset for remote sensing applications. It was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) in Pavia, Italy. The dataset has a size of 610 × 340 with 103 spectral bands and a spatial resolution of 1.3 m/pixel. The image covers an urban area, including various land cover classes, such as asphalt, meadows, trees, and bare soil. Fig. 4(a) and (b) represents the false-color image along with its corresponding ground truth.
3) The Houston hyperspectral dataset is also a commonly used dataset for remote sensing and HSI analysis. It was collected over an urban area in Houston, TX, USA, using the AVIRIS sensor. The Houston dataset contains 349 × 1905 pixels with 144 spectral bands and a spatial resolution of approximately 2 m/pixel. It covers an urban environment with various land cover classes, including buildings, roads, vegetation, and water bodies. Fig. 5(a) and (b) presents the false-color image along with its corresponding ground truth.
4) The WHU-Hi-HanChuan is a new famous hyperspectral dataset for remote sensing and HSI analysis. It was collected in Hanchuan, Hubei, China, using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform. The WHU-Hi-HanChuan dataset contains 1217 × 303 pixels with 274 spectral bands from 400 to 1000 nm. Fig. 6(a) and (b) presents the true-color image along with its corresponding ground truth.

B. Experimental Settings

In this section, based on the three classic datasets introduced above, we compared methods DPRN [23], ASSMN [45], and SSFTT [46] based on CNN, methods Funet-C, MDGCN [33], and FDGC [28] based on GCN, the dual-branch approach DSGSF [24], and morphFormer [47] and GraphGST [48] based on Transformer. The proposed LPDGCN method is implemented using the TensorFlow framework. In our approach, we set the number of hidden layers in GCN to 20 and the number of epochs to 5000. In the comparative experiments, we selected 15 labeled pixels from each class for training and then evaluated the classification performance on all unlabeled samples as the test set, as shown in Table I for the numbers of training and testing samples. Furthermore, to mitigate the randomness introduced by sampling, we repeated each method ten times and calculated their averages as the final results. Finally, we used overall accuracy (OA), average accuracy (AA), and the kappa coefficient (KC) to assess the performance of all methods in HSI classification. The closer the metrics are to 1, the better the classification performance of the model.

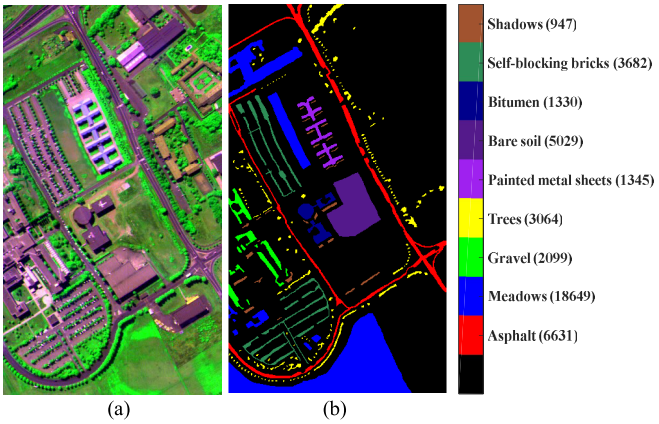

Fig. 4. PaviaU data: (a) false-color image and (b) ground-truth image (bands 52, 99, and 4 are red, green, and blue, respectively).

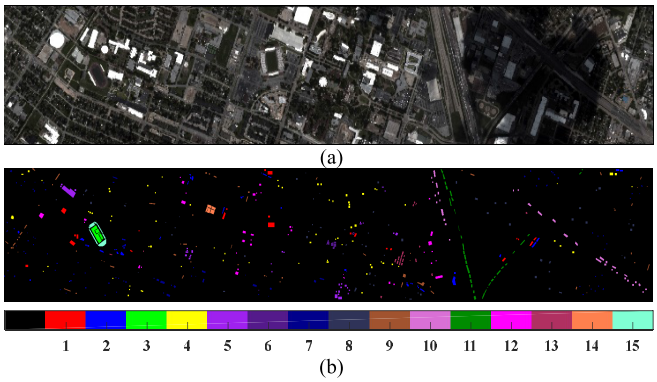

Fig. 5. Houston data: (a) false-color image (bands 45, 38, and 32 are red, green, and blue, respectively) and (b) ground-truth image.

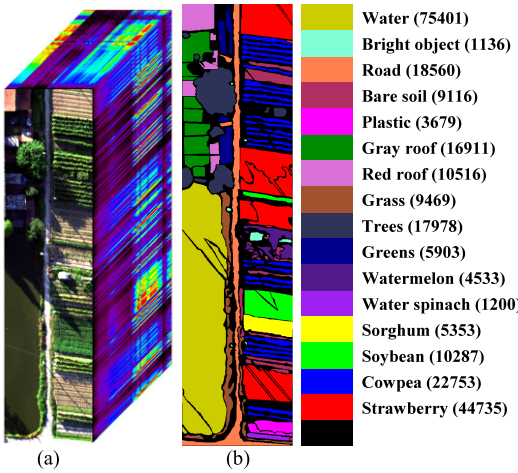

Fig. 6. WHU-Hi-HanChuan data: (a) true-color image and (b) ground-truth image.

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TABLE I
DETAILED NUMBER OF TRAINING AND TESTING SAMPLES ON INDIAN PINES, PAVIAU, HOUSTON, AND WHU-HI-HANCHUAN DATASETS

Indian Pines				PaviaU			
Class	Name	Train	Test	Class	Name	Train	Test
1	Alfalfa	15	31	1	Asphalt	15	2150
2	Corn-notill	15	1413	2	Meadows	15	2694
3	Corn-mintill	15	815	3	Gravel	15	1791
4	Corn	15	222	4	Trees	15	833
5	Grass-pasture	15	468	5	Painted metal sheets	15	1330
6	Grass-trees	15	715	6	Bare Soil	15	4942
7	Grass-pasture-mowed	15	13	7	Bitumen	15	1315
8	Hay-windrowed	15	463	8	Self-Blocking Bricks	15	2009
9	Oats	15	5	9	Shadows	15	841
10	Soybean-notill	15	957		Total	135	17905
11	Soybean-mintill	15	2440				
12	Soybean-clean	15	578				
13	Wheat	15	190				
14	Woods	15	1250				
15	Buildings-Grass-Trees-Drives	15	371				
16	Stone-Steel-Towers	15	79				
	Total	240	10619				
Houston				WHU-Hi-HanChuan			
Class	Name	Train	Test	Class	Name	Train	Test
1	Healthy grass	15	1236	1	Strawberry	15	44720
2	Stressed grass	15	1239	2	Cowpea	15	22738
3	Synthetic grass	15	682	3	Soybean	15	10272
4	Tree	15	1229	4	Sorghum	15	5338
5	Soil	15	1227	5	Water spinach	15	1185
6	Water	15	310	6	Watermelon	15	4518
7	Residential	15	1253	7	Greens	15	5888
8	Commercial	15	1229	8	Trees	15	17963
9	Road	15	1237	9	Grass	15	9454
10	Highway	15	1212	10	Red roof	15	10501
11	Railway	15	1220	11	Gray roof	15	16896
12	Parking lot 1	15	1218	12	Plastic	15	3664
13	Parking lot 2	15	454	13	Bare soil	15	9101
14	Tennis court	15	413	14	Road	15	18545
15	Running track	15	645	15	Bright object	15	1121
	Total	225	14804	16	Water	15	75386
					Total	240	257290

C. Experimental Results

The overall classification performance of all the methods used in the comparative experiments on the Indian Pines, PaviaU, and Houston datasets is shown in Tables II–IV. Table II and Fig. 7 present the overall classification performance and classification maps of different methods on the Indian Pines dataset. Among these methods, the LPDGCN method achieves an overall classification accuracy of over 90%, outperforming other mainstream methods. Table III and Fig. 8 display the overall classification performance and classification maps for different methods on the PaviaU dataset. In this case, the LPDGCN method exhibits higher accuracy in terms of OA, AA, and KC compared to other methods. Table IV and Fig. 9 represent the performance of different models on the Houston dataset, where the LPDGCN method still outperforms other methods with an OA of over 84%. Table V and Fig. 10 show the classification performance of different models on the WHU-Hi-HanChuan dataset, where the LPDGCN method achieves an OA of over 85%, representing a significant improvement compared to other methods. In addition, across all three datasets, LPDGCN exhibits the least fluctuation in accuracy, demonstrating the best stability. From several tables, it can be observed that the LPDGCN achieves the highest classification performance in terms of OA, AA, and KC. Most models primarily focus on the ability to extract pixel-level feature information to compensate for deficiencies in superpixel details. What sets LPDGCN apart is its strong emphasis on improving the quality of the input graph and addressing the issue of the spatial distribution of labeled nodes after input, which enhances the quality of the extracted information and addresses the problem of redundancy in interclass information, resulting in more accurate classification results.

D. Analysis of Different Training Samples

To further validate the effectiveness of the LPDGCN method, we conducted multiple experiments with varying numbers of training samples. Fig. 11 demonstrates the experimental results on four datasets of the LPDGCN method and several other methods. We choose 5, 10, 15, 20, 25, and 30 training samples extracted from each class in the Indian Pines, PaviaU, Houston, and WHU-Hi-HanChuan datasets. The experimental results indicate that the LPDGCN method consistently outperforms other comparative methods across various numbers of training samples. From the graphs, it can be observed that the LPDGCN method consistently outperforms other comparative methods across various numbers of training samples. From the graphs, it can be observed that even

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT COMPARED METHODS AND LPDGCN METHOD ON INDIAN PINES DATASET

Class	DPRN	ASSMN	SSFTT	morphFormer	GraphGST	MDGCN	Funet-C	FDGC	DSGSF	LPDGCN
1	**100**(0)	**100**(0)	**100**(0)	**100**(0)	99.03(1.48)	95.48(5.04)	94.84(4.83)	39.67(11.66)	76.15(6.93)	94.84(3.87)
2	58.62(11.46)	63.12(9.98)	61.61(4.93)	70.04(16.14)	**80.84**(7.45)	77.73(6.89)	43.08(6.97)	76.83(8.37)	73.07(8.46)	78.49(5.21)
3	63.29(8.50)	72.85(9.18)	85.34(3.03)	79.74(6.72)	84.43(7.84)	85.10(6.72)	50.63(6.80)	77.68(5.23)	70.71(8.34)	**86.37**(8.86)
4	84.82(8.42)	95.95(4.37)	**97.88**(1.12)	89.74(8.63)	95.63(2.60)	96.35(3.89)	69.05(6.22)	67.13(17.29)	56.63(16.80)	95.95(3.54)
5	75.49(7.95)	81.24(8.47)	**99.34**(0.72)	85.94(7.12)	88.06(7.23)	89.00(4.44)	80.49(4.49)	87.88(6.92)	86.13(4.95)	87.35(2.73)
6	86.13(6.38)	84.62(6.56)	**99.01**(0.56)	95.76(3.71)	92.98(4.65)	93.03(7.53)	90.88(4.51)	93.16(2.74)	96.50(2.57)	95.16(2.58)
7	**100**(0)	**100**(0)	**100**(0)	98.89(2.22)	96.92(3.77)	93.85(8.97)	99.23(2.31)	36.76(24.37)	47.67(15.88)	99.23(2.31)
8	97.84(1.39)	98.92(3.17)	99.44(0.70)	**100**(0)	**100**(0)	99.78(0.43)	88.98(6.12)	99.11(1.81)	98.28(1.53)	99.63(1.10)
9	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	**100**(0)	11.42(9.42)	35.06(23.32)	**100**(0)
10	72.57(9.52)	71.54(9.25)	88.44(1.54)	**91.08**(3.30)	87.60(1.01)	84.72(6.45)	62.29(6.13)	78.72(6.23)	73.70(8.06)	88.36(6.36)
11	67.30(5.13)	76.70(7.22)	85.64(2.15)	67.85(6.69)	75.14(4.79)	84.57(4.08)	56.07(5.70)	**91.43**(4.72)	90.22(6.58)	90.07(2.38)
12	60.31(6.39)	77.46(6.62)	73.18(3.35)	80.94(3.27)	87.01(5.07)	87.08(5.39)	54.36(5.49)	65.97(8.46)	64.31(13.14)	**87.11**(6.60)
13	99.42(0.60)	99.47(0.78)	**100**(0)	99.44(0.28)	99.32(0.34)	**100**(0)	99.37(0.21)	78.49(14.21)	94.84(6.33)	98.68(1.32)
14	92.26(4.78)	84.46(7.57)	98.43(0.83)	92.37(7.60)	91.77(7.09)	95.14(4.84)	82.33(4.65)	95.79(2.98)	**98.49**(0.91)	98.38(3.26)
15	84.83(4.73)	95.18(5.49)	94.61(2.42)	92.55(7.54)	98.38(1.35)	96.28(5.69)	48.06(10.67)	79.89(10.06)	78.66(8.88)	**98.46**(1.19)
16	97.05(3.54)	99.10(1.99)	96.79(1.31)	99.88(0.36)	**100**(0)	96.92(1.83)	97.05(3.89)	67.17(8.97)	77.92(12.50)	95.51(2.09)
OA	74.15(2.80)	78.63(3.23)	86.63(0.83)	81.91(2.15)	85.76(1.23)	87.78(1.39)	64.05(2.16)	82.01(1.56)	81.16(3.33)	**90.20**(1.20)
AA	83.74(1.60)	87.54(2.10)	92.48(0.43)	90.26(1.48)	92.32(0.92)	92.19(1.15)	71.57(1.19)	71.75(2.48)	76.15(3.89)	**93.35**(0.97)
KC	70.72(3.19)	75.78(3.61)	84.80(0.93)	79.68(2.46)	83.89(1.37)	86.12(1.56)	59.43(2.29)	79.68(1.74)	78.69(3.77)	**88.83**(1.38)

TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT COMPARED METHODS AND LPDGCN METHOD ON PAVIAU DATASET

Class	DPRN	ASSMN	SSFTT	morphFormer	GraphGST	MDGCN	Funet-C	FDGC	DSGSF	LPDGCN
1	85.21(4.48)	72.99(8.14)	88.48(8.00)	83.17(5.60)	89.20(5.31)	79.58(5.86)	67.10(10.59)	89.70(3.01)	**94.28**(2.92)	83.86(3.61)
2	89.99(5.86)	77.00(5.93)	91.94(7.01)	64.26(14.73)	75.93(4.84)	91.48(4.13)	62.98(9.84)	98.41(0.90)	**96.06**(3.37)	93.28(3.08)
3	64.46(11.80)	60.31(6.03)	90.39(8.34)	92.55(3.52)	**92.98**(6.24)	88.18(8.49)	39.68(11.87)	87.01(7.20)	77.16(11.80)	92.46(3.19)
4	85.51(8.13)	91.47(2.23)	79.51(4.51)	**94.53**(4.36)	93.94(3.08)	88.46(3.63)	84.14(5.64)	66.62(7.87)	74.54(15.05)	89.14(3.92)
5	99.53(0.57)	95.89(5.74)	99.36(0.71)	99.94(0.03)	**99.95**(0.02)	96.71(2.76)	99.25(0.11)	96.27(2.98)	96.22(2.29)	97.38(3.84)
6	66.06(11.86)	69.81(9.63)	84.44(7.96)	71.35(9.87)	71.48(8.61)	91.45(3.78)	81.97(7.79)	79.00(10.28)	76.66(14.25)	**95.83**(3.24)
7	75.05(13.89)	89.79(5.96)	**99.34**(1.15)	94.41(2.21)	94.44(3.47)	90.12(4.30)	48.21(22.56)	89.60(9.98)	68.81(15.96)	86.49(6.88)
8	**84.42**(8.41)	79.39(10.26)	75.06(15.76)	83.03(9.90)	84.82(5.84)	82.66(5.76)	49.81(15.04)	76.97(7.93)	81.31(5.81)	81.92(8.65)
9	98.42(1.12)	99.41(0.70)	95.24(8.12)	88.83(4.07)	92.63(3.42)	86.48(6.65)	**100**(0)	79.49(8.50)	95.69(3.95)	84.07(6.01)
OA	84.41(2.92)	77.43(2.49)	88.01(2.61)	75.78(4.67)	82.04(2.67)	88.51(1.90)	72.64(4.62)	87.57(2.16)	86.65(3.04)	**90.52**(1.41)
AA	83.19(2.90)	81.78(1.62)	88.86(1.62)	85.78(0.92)	88.37(1.16)	88.35(1.29)	65.49(2.33)	84.78(1.82)	84.52(2.99)	**89.38**(1.28)
KC	79.35(3.74)	71.04(2.88)	84.24(3.26)	69.89(4.85)	77.02(3.17)	84.97(2.40)	65.20(5.29)	83.89(2.71)	82.76(3.91)	**87.57**(1.78)

TABLE IV
CLASSIFICATION RESULTS (%) OF DIFFERENT COMPARED METHODS AND LPDGCN METHOD ON HOUSTON DATASET

Class	DPRN	ASSMN	SSFTT	morphFormer	GraphGST	MDGCN	Funet-C	FDGC	DSGSF	LPDGCN
1	**94.97**(4.23)	88.44(6.62)	93.15(1.91)	88.67(5.42)	89.86(6.26)	90.79(5.74)	92.90(5.69)	87.87(5.20)	86.69(7.30)	89.71(3.16)
2	88.20(3.70)	74.24(10.47)	79.72(3.84)	93.01(4.78)	92.97(4.10)	88.51(4.68)	**97.43**(3.78)	86.02(8.34)	89.75(8.04)	88.57(5.21)
3	97.29(2.29)	87.84(16.07)	**97.60**(1.25)	93.51(4.32)	93.86(4.71)	96.19(2.65)	98.11(1.36)	95.80(3.74)	81.51(20.55)	96.03(2.56)
4	90.34(4.15)	86.52(11.29)	88.98(2.61)	86.20(4.35)	86.06(4.81)	90.46(2.78)	**94.05**(2.80)	77.98(7.88)	91.40(8.12)	92.04(2.81)
5	98.28(2.85)	87.38(7.58)	**99.81**(0.34)	99.37(0.52)	99.67(0.35)	98.89(1.85)	99.05(0.70)	95.99(2.73)	94.96(2.64)	99.70(0.30)
6	82.48(5.21)	87.55(4.42)	**99.25**(0.92)	92.89(5.41)	90.86(6.49)	88.87(8.84)	89.58(5.02)	82.05(9.44)	93.77(6.14)	90.32(6.13)
7	74.41(8.61)	70.80(16.13)	68.46(5.03)	67.11(7.80)	68.69(9.76)	78.84(8.99)	71.12(6.58)	65.38(7.22)	76.97(12.65)	**82.72**(6.02)
8	64.17(5.60)	45.31(11.20)	68.58(2.85)	66.98(11.62)	65.15(8.55)	70.72(7.53)	67.88(6.09)	**90.43**(6.42)	74.81(10.82)	72.13(6.58)
9	72.34(6.54)	78.52(7.31)	67.03(6.76)	68.93(9.26)	72.66(7.47)	83.51(4.55)	70.57(5.81)	76.17(7.26)	71.50(6.58)	**83.59**(5.11)
10	65.87(11.74)	30.21(20.13)	**85.04**(5.37)	75.48(12.84)	79.11(9.03)	73.26(9.06)	71.20(9.16)	81.80(6.79)	67.63(7.63)	75.51(7.59)
11	72.22(7.39)	55.14(14.76)	79.32(5.89)	72.18(7.98)	71.91(10.75)	64.03(10.64)	63.70(6.64)	**87.23**(8.20)	75.30(12.12)	73.25(7.05)
12	68.92(8.51)	34.46(11.42)	63.43(11.64)	73.13(9.77)	69.00(9.88)	66.75(7.23)	80.11(7.08)	**92.48**(4.95)	61.70(7.27)	74.86(7.14)
13	74.78(18.74)	38.37(17.68)	**86.94**(4.21)	82.59(4.51)	83.51(5.30)	76.38(10.19)	68.37(5.49)	64.85(8.23)	52.47(19.61)	78.65(10.67)
14	95.26(6.93)	94.72(6.33)	**99.87**(0.38)	98.68(1.18)	99.27(1.20)	99.59(0.41)	94.46(4.72)	95.74(4.10)	88.45(6.16)	99.70(0.27)
15	98.90(1.22)	94.60(4.29)	**99.95**(0.12)	99.55(0.28)	99.52(0.28)	97.55(1.39)	99.61(0.39)	87.16(9.88)	94.20(3.94)	96.85(1.77)
OA	81.11(1.77)	68.01(4.10)	81.91(1.91)	81.24(1.64)	81.64(1.48)	82.28(1.06)	82.62(1.52)	82.01(1.56)	78.57(3.47)	**84.59**(0.64)
AA	82.56(1.95)	70.27(4.07)	85.14(1.55)	83.89(1.52)	84.14(1.06)	84.29(0.84)	78.63(1.19)	71.75(2.48)	80.07(3.61)	**86.24**(1.01)
KC	79.58(1.91)	65.53(4.37)	80.44(2.07)	79.73(1.77)	80.16(1.59)	80.84(1.14)	81.21(1.65)	79.68(1.74)	76.85(3.75)	**83.34**(0.69)

with a limited number of samples, LPDGCN still achieves satisfactory performance, while most other methods perform poorly when there are only five samples.

However, as the number of samples increases, other methods show significant performance improvements. This indicates that LPDGCN has a more stable performance compared to other models across changes in the number of training samples. In contrast, most other methods require a larger number of training samples, indicating the need for obtaining richer graph information to achieve good results. The main reason for the result is that the LPDGCN method reduces the negative impact of labeled nodes and enhances the positive impact of labeled nodes by addressing the imbalance in the position distribution. As a result, LPDGCN demonstrates lower sensitivity to changes in the number of labeled nodes compared to other models, resulting in a more stable performance.

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TABLE V
CLASSIFICATION RESULTS (%) OF DIFFERENT COMPARED METHODS AND LPDGCN METHOD ON WHU-HI-HANCHUAN DATASET

Class	DPRN	ASSMN	SSFTT	morphFormer	GraphGST	MDGCN	FDGC	DSGSF	LPDGCN
1	73.22(4.59)	65.93(12.05)	85.28(9.26)	78.17(7.28)	81.61(6.87)	85.07(8.19)	**95.98**(1.69)	95.09(2.00)	88.26(4.61)
2	47.13(10.68)	76.87(9.48)	68.11(7.66)	65.53(7.47)	69.98(5.32)	44.55(10.48)	**88.10**(7.28)	81.81(8.13)	59.34(6.25)
3	73.47(8.13)	68.68(12.54)	81.19(8.19)	77.77(8.63)	80.22(7.83)	**91.95**(2.94)	83.84(6.97)	79.21(8.96)	86.65(5.28)
4	93.85(2.78)	92.53(6.04)	**98.71**(0.62)	96.16(1.62)	96.23(2.45)	95.11(1.17)	86.63(8.60)	87.51(5.93)	95.83(1.47)
5	87.55(6.86)	96.00(5.24)	90.71(5.87)	**98.58**(1.16)	97.92(3.17)	87.74(5.15)	55.71(17.32)	54.22(21.85)	86.57(8.43)
6	38.17(7.26)	43.51(11.53)	56.56(15.52)	66.52(5.09)	**76.13**(5.80)	63.57(6.63)	51.71(8.76)	30.69(5.51)	69.37(10.37)
7	82.36(12.75)	85.30(8.96)	91.83(3.60)	89.71(4.51)	88.43(5.58)	92.33(2.14)	61.18(12.80)	64.19(12.69)	**93.42**(2.34)
8	51.39(7.16)	50.72(8.23)	64.91(3.70)	63.03(10.28)	66.87(13.21)	63.04(4.98)	**88.48**(6.88)	75.94(8.69)	67.58(3.64)
9	42.51(6.11)	49.25(12.77)	72.50(6.56)	58.16(11.20)	67.63(11.11)	68.55(3.64)	64.08(7.75)	55.47(15.14)	**72.85**(8.06)
10	87.65(5.76)	49.70(15.36)	89.11(11.91)	**97.64**(1.40)	96.56(0.95)	94.12(3.76)	92.32(5.34)	86.68(7.39)	93.85(3.62)
11	71.38(13.92)	52.51(22.90)	88.09(5.97)	87.53(7.88)	91.09(5.49)	**97.82**(0.96)	84.63(7.21)	69.61(13.12)	95.13(3.44)
12	66.53(16.48)	72.16(15.32)	67.68(7.03)	82.28(4.20)	83.21(8.54)	**84.72**(4.19)	43.09(10.98)	53.11(7.53)	84.64(5.53)
13	53.36(4.55)	53.29(7.28)	49.37(7.97)	60.53(9.52)	70.51(6.68)	**75.23**(7.29)	44.76(6.97)	54.25(10.10)	72.33(9.76)
14	71.10(4.35)	76.17(6.77)	75.83(9.12)	69.61(3.81)	69.77(6.92)	71.64(5.20)	78.99(12.22)	**83.51**(8.52)	74.18(6.18)
15	72.49(5.01)	79.53(6.40)	**92.07**(3.36)	74.65(5.71)	80.44(5.94)	74.45(8.80)	34.63(13.91)	43.96(15.48)	79.59(9.12)
16	97.07(1.19)	89.89(8.94)	96.26(2.42)	95.62(4.66)	94.69(5.04)	95.66(5.00)	98.62(2.18)	**99.26**(0.52)	97.90(1.44)
OA	74.86(0.98)	71.99(4.03)	83.04(2.31)	81.10(2.07)	83.26(2.24)	82.57(1.83)	82.95(1.54)	80.26(3.38)	**85.40**(1.41)
AA	69.33(2.23)	68.88(4.87)	79.26(2.09)	78.84(1.05)	81.95(2.16)	80.35(1.37)	72.05(2.25)	69.66(3.83)	**82.34**(1.81)
KC	70.93(1.07)	67.46(4.71)	80.28(2.65)	78.11(2.27)	80.62(2.53)	79.84(2.04)	80.27(1.77)	77.26(3.80)	**83.03**(1.62)

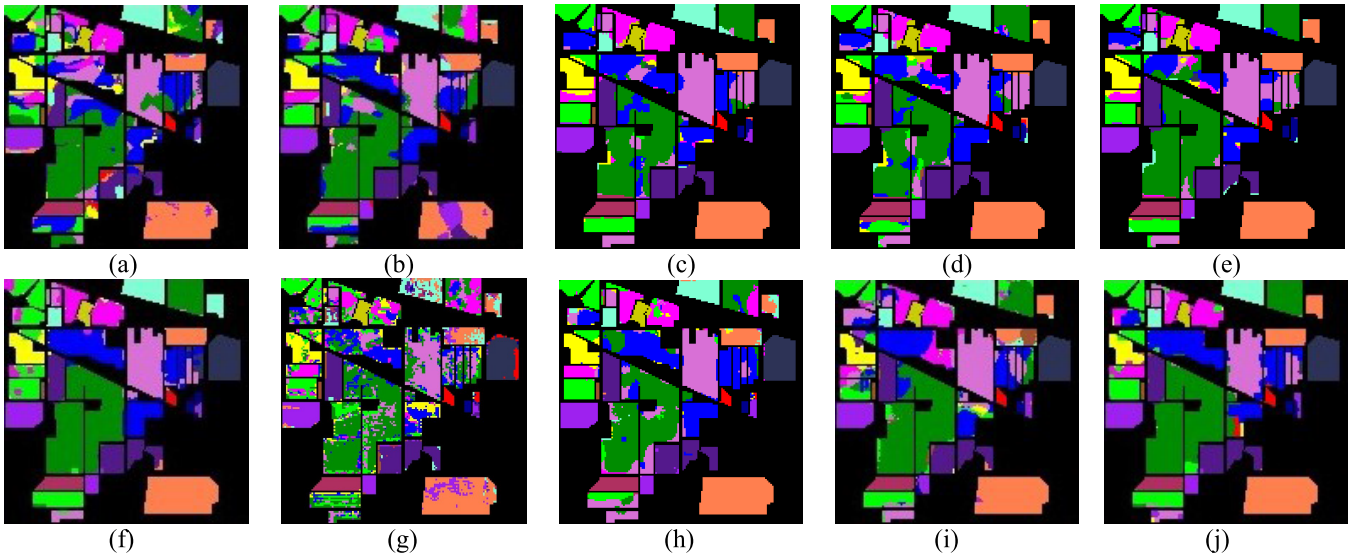

Fig. 7. Classification maps of different methods on Indian Pines dataset. (a) DPRN. (b) ASSMN. (c) SSFTT. (d) morphFormer. (e) GraphGST. (f) MDGCN. (g) Funet-C. (h) FDGC. (i) DSGSF. (j) LPDGCN.

E. Parameter Analysis

The method proposed in this article mainly consists of seven parameters, including λ_1, λ_2, and λ_3 for the graph reconstruction part, and λ, α, β, and γ for the weighting labeled nodes part of the LPDGCN model. Among these, λ_1 and λ_2, respectively, control the degree of sparsity and low rankness in constraining the reconstructed adjacency matrix, while λ_3 regulates the similarity between features of the two nodes connected by edges eliminated during the reconstruction process. Parameter α controls the magnitude of the restart probability used in computing the influence distribution of labeled nodes through the PPR matrix, and β and γ govern the impact of weights of labeled nodes. Among these parameters, the parameters α and λ have a larger impact on the model compared to the others. The adjustment of α determines the results of topological position calculations for labeled nodes within the graph structure. Overall, this parameter significantly influences the determination of the topological position of labeled nodes within the graph structure. In this article, the parameter α is set to 0.15. Fig. 12 illustrates the classification results when adjusting parameter α in the range of 0.05–0.95 on the four datasets. The parameter λ determines the extent of the influence of the position-reweighted component in the LPDGCN model during training. In this article, the parameter λ is set to 1. Fig. 13 shows the different classification results on the four datasets when the parameter λ is set from 0.1 to 10. The results indicate that the impact of the position-reweighted component plays a significant role in the classification results. During this process, the LPDGCN method initially rises to a peak and then continues to decrease, indicating that the parameter choice is within a reasonable range.

F. Ablation Study

The optimization process of the proposed LPDGCN in this article can be primarily divided into two parts: the reconstruction of low-rank and sparse graph structures and the

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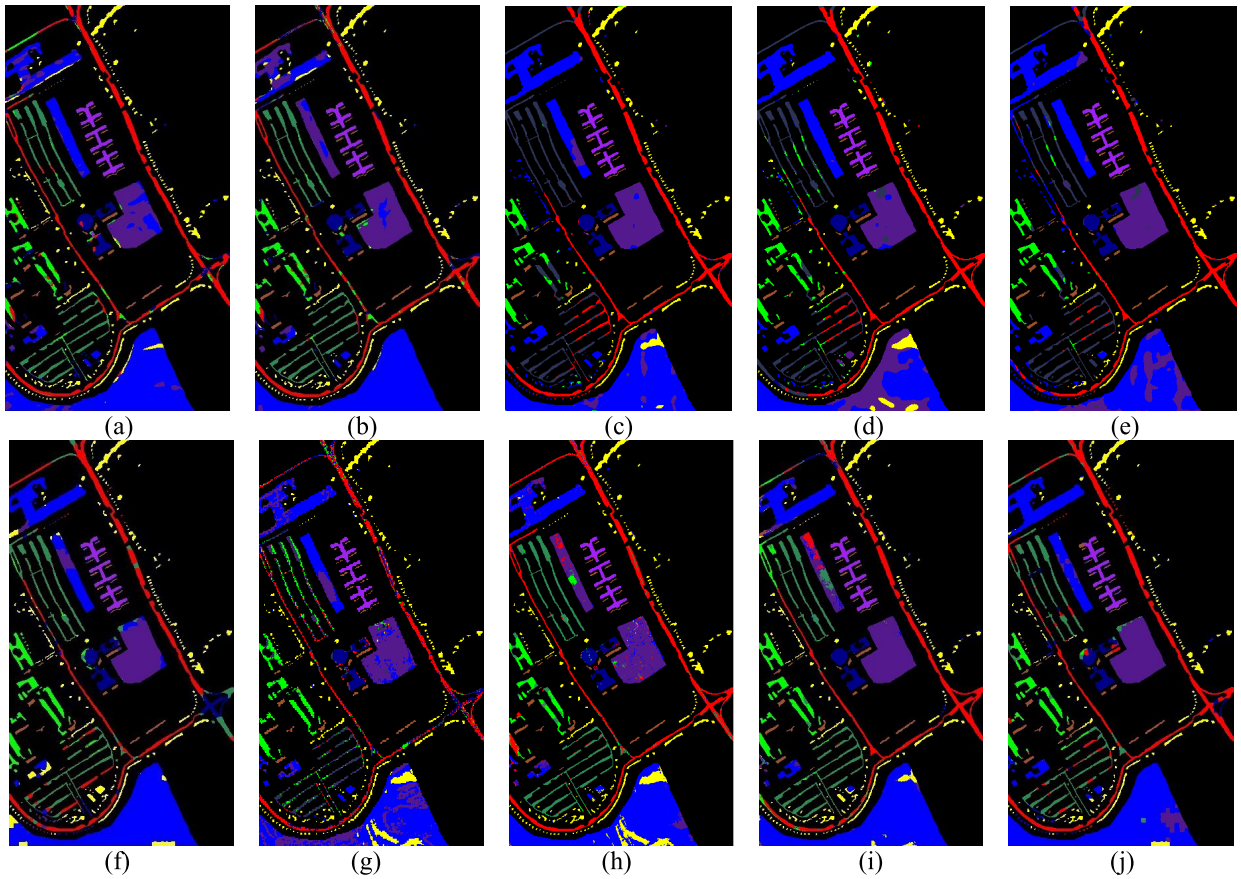

Fig. 8. Classification maps of different methods on PaviaU dataset. (a) DPRN. (b) ASSMN. (c) SSFTT. (d) morphFormer. (e) GraphGST. (f) MDGCN. (g) Funet-C. (h) FDGC. (i) DSGSF. (j) LPDGCN.

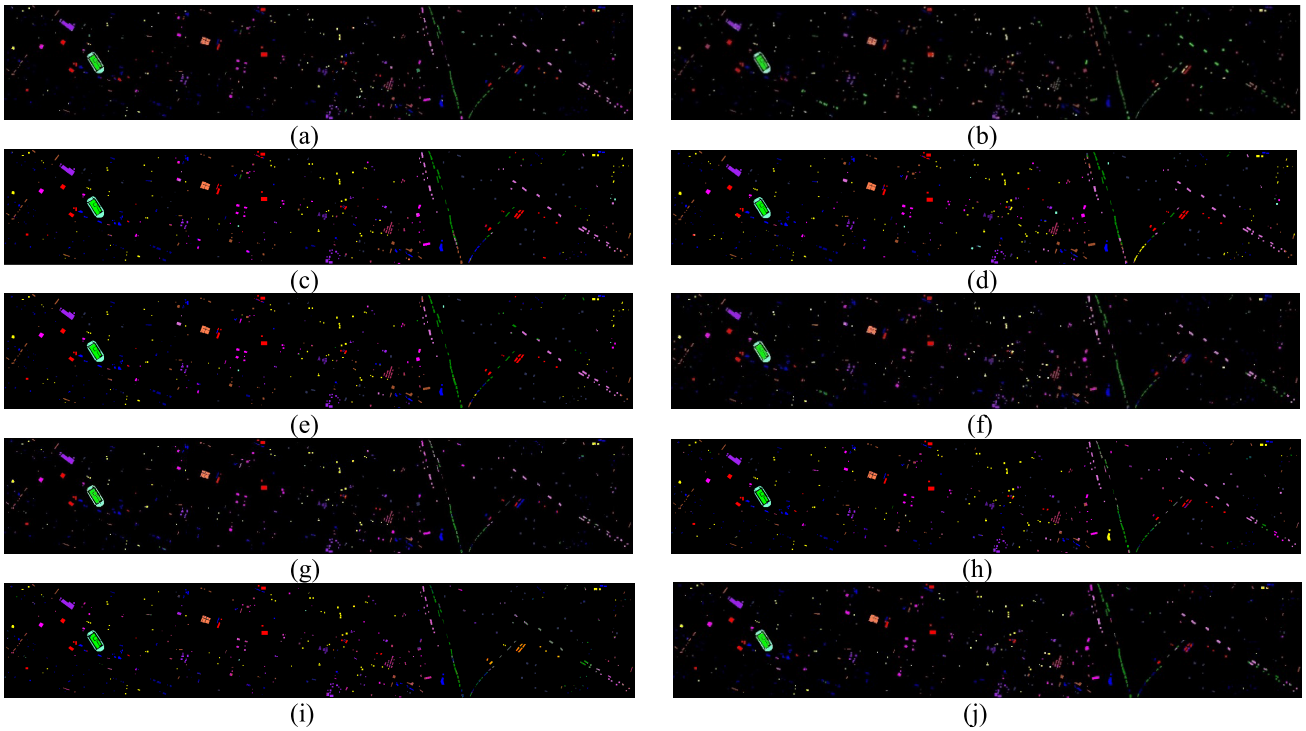

Fig. 9. Classification maps of different methods on Houston dataset. (a) DPRN. (b) ASSMN. (c) SSFTT. (d) morphFormer. (e) GraphGST. (f) MDGCN. (g) Funet-C. (h) FDGC. (i) DSGSF. (j) LPDGCN.

weighting of labeled nodes. We conducted experiments on the Indian Pines, PaviaU, and Houston datasets to demonstrate the effectiveness of these two parts individually, as well as their combined enhancement. Table VI presents the results of ablation experiments under different training sample sizes. LPD-W represents only weighting labeled nodes, while LPD-R

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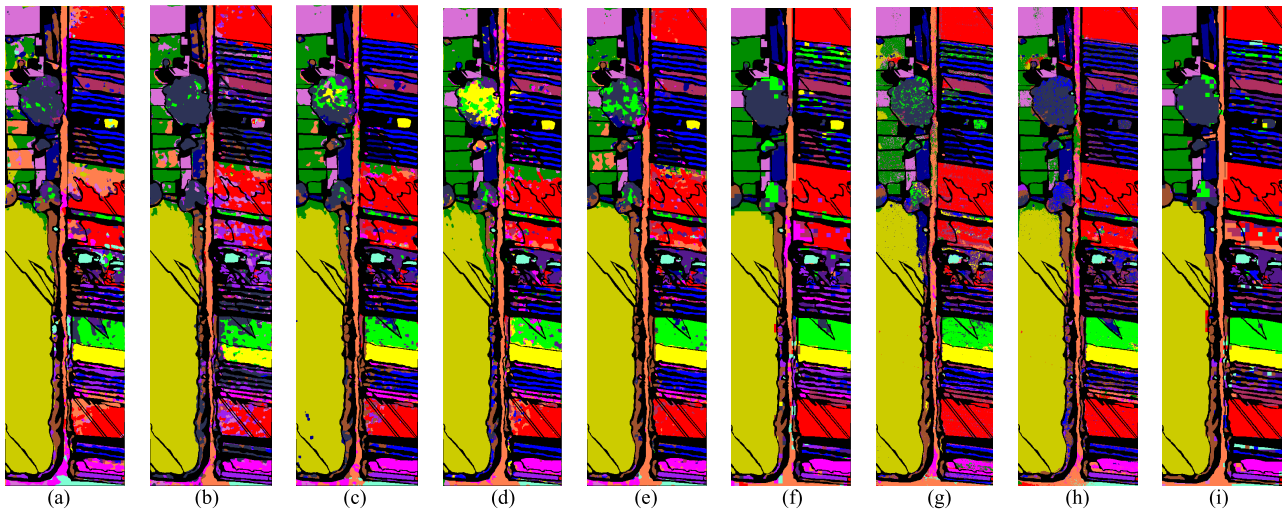

Fig. 10. Classification maps of different methods on WHU-Hi-HanChuan dataset. (a) DPRN. (b) ASSMN. (c) SSFTT. (d) morphFormer. (e) GraphGST. (f) MDGCN. (g) FDGC. (h) DSGSF. (i) LPDGCN.

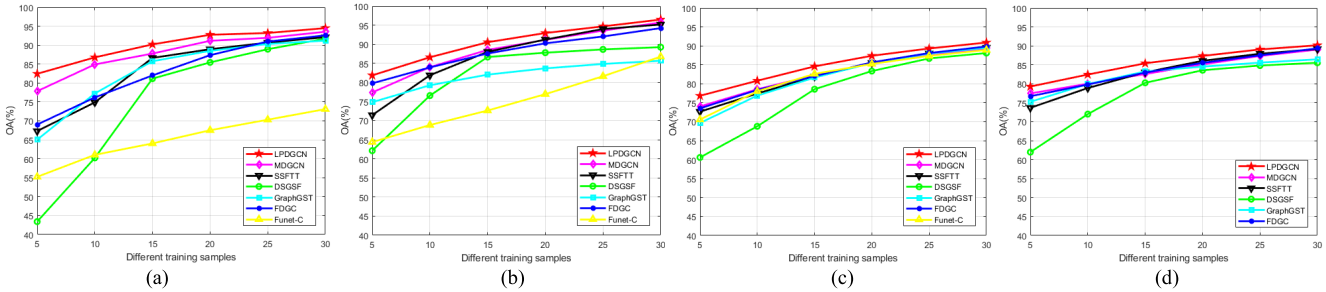

Fig. 11. OAs of different training samples on different methods and LPDGCN method for four HSI datasets. (a) Indian Pines. (b) PaviaU. (c) Houston. (d) WHU-Hi-HanChuan.

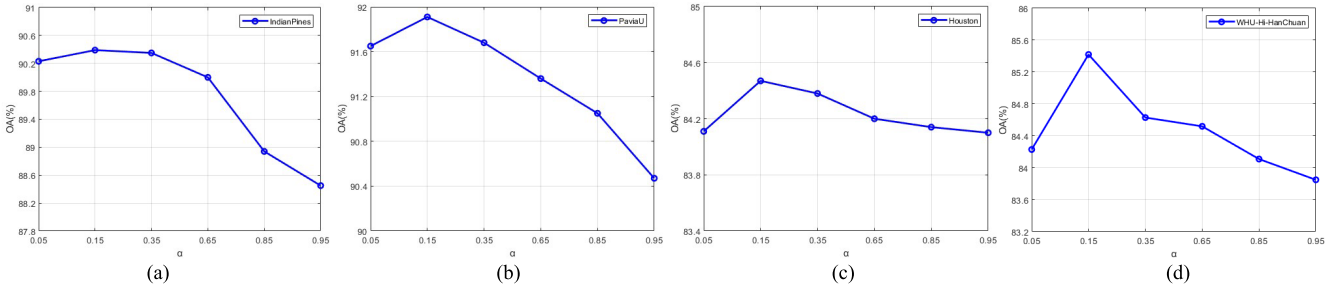

Fig. 12. OAs of different parameter α on LPDGCN method for four HSI datasets. (a) Indian Pines. (b) PaviaU. (c) Houston. (d) WHU-Hi-HanChuan.

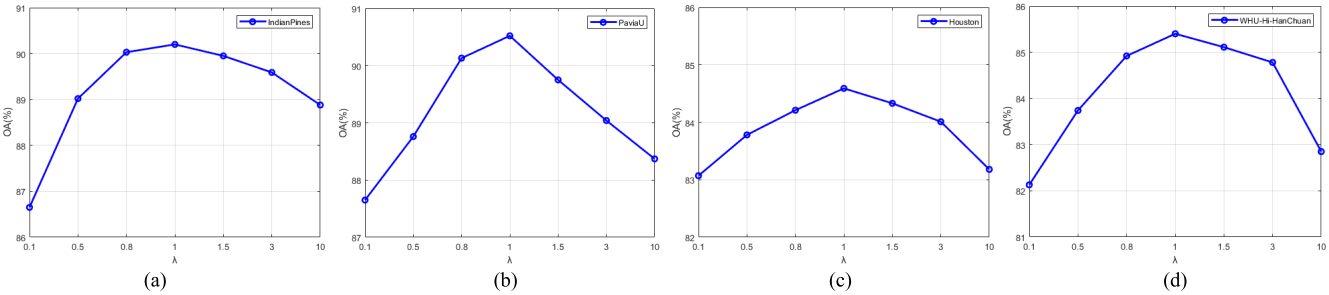

Fig. 13. OAs of different parameter λ on LPDGCN method for four HSI datasets. (a) Indian Pines. (b) PaviaU. (c) Houston. (d) WHU-Hi-HanChuan.

represents only rebuilding the superpixel graph structure. The results indicate that both the reconstruction and weighting parts significantly contribute to improving classification performance. It is worth noting that the performance of LPDGCN is not simply the sum of improvements from both parts, and it handles the essential imbalanced position distributions in

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TABLE VI
ABLATION EXPERIMENTS' OAS (%) OF DIFFERENT TRAINING SAMPLES ON FOUR DATASETS

Indian Pines						
Samples	5	10	15	20	25	30
LPDGCN	**82.44**	**86.77**	**90.20**	**92.74**	**93.21**	**94.50**
LPD-R	80.42	85.28	89.26	91.85	93.05	94.16
LPD-W	80.22	85.84	88.64	91.69	92.03	93.88
PaviaU						
Samples	5	10	15	20	25	30
LPDGCN	**81.35**	**86.58**	**90.52**	**92.93**	**94.67**	**96.45**
LPD-R	79.38	85.63	89.08	91.82	93.98	96.02
LPD-W	79.52	85.44	89.13	91.76	94.07	95.89
Houston						
Samples	5	10	15	20	25	30
LPDGCN	**76.84**	**80.81**	**84.59**	**87.39**	**89.31**	**90.88**
LPD-R	75.96	80.23	83.69	86.96	88.82	90.18
LPD-W	75.89	80.31	83.88	86.71	88.54	90.34
WHU-Hi-HanChuan						
Samples	5	10	15	20	25	30
LPDGCN	**80.28**	**83.44**	**85.40**	**87.35**	**89.06**	**90.16**
LPD-R	79.33	82.25	84.49	86.66	88.65	89.66
LPD-W	78.94	81.88	84.04	86.91	88.41	89.43

superpixel-level GCN-based methods achieving better classification performance. In fact, the sparse, low-rank as well as feature smoothing constraints in the reconstructed graph alleviate the influence of redundant interclass connections, obtaining a qualified graph structure. Next, the qualified graph structure further promotes the location determination of labeled nodes, making the weighting of label nodes operation crucial.

IV. CONCLUSION

In this article, we propose an LPDGCN method that addresses the issue of imbalanced position distributions of labeled nodes, achieving good classification performance. The LPDGCN is mainly consisting two modules that are the optimization of graph structure and the reweighted strategy in position distribution. The optimization of graph structure can be obtained by the sparse, low-rank as well as feature smoothness restrictions for removing redundant interclass connections. The reweighted strategy in position distribution is to weaken the influence of labeled superpixels at class boundaries and strengthen the influence of labeled superpixels at class centers based on the determination of the position distributions of labeled superpixels from the above topology connectivity. Finally, we evaluate the LPDGCN method on four public HSI datasets, demonstrating its superiority over other advanced classification methods in terms of three metrics, i.e., OA, AA, and KC.

In addition, there is still room for further consideration regarding the proposed method in this article. For instance, although the LPDGCN method proposed in this article effectively addresses the impact of the position distribution of labeled superpixels on model training, there remains the challenge of accurately determining the positional relationships. Future work will involve the generation of high-quality labeled superpixels using generative models, which will result in a more uniform and regular distribution of labeled superpixels, alleviating the issue of the position distribution of labeled superpixels.

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